MA148: CALCULUS WITH APPLICATIONS

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Applied Calculus (Callaway, Hoffman, Lippman) - Butler CC Remix

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Introduction

Introduction

A Preview of Calculus

Calculus was first developed more than three hundred years ago by Sir Isaac Newton and Gottfried Leibniz to help them describe and understand the rules governing the motion of planets and moons. Since then, thousands of other men and women have refined the basic ideas of calculus, developed new techniques to make the calculations easier, and found ways to apply calculus to problems besides planetary motion. Perhaps most importantly, they have used calculus to help understand a wide variety of physical, biological, economic and social phenomena and to describe and solve problems in those areas.

Part of the beauty of calculus is that it is based on a few very simple ideas. Part of the power of calculus is that these simple ideas can help us understand, describe, and solve problems in a variety of fields.

About this book

Chapter 1 Review contains review material that you should recall before we begin calculus.

Chapter 2 The Derivative builds on the precalculus idea of the slope of a line to let us find and use rates of change in many situations.

Chapter 3 The Integral builds on the precalculus idea of the area of a rectangle to let us find accumulated change in more complicated and interesting settings.

Chapter 4 Functions of Two Variables extends the calculus ideas of chapter 2 to functions of more than one variable.

Supplements

An online course framework is available on MyOpenMath.com for this book. The course framework features:

- Links to individual sections of the e-text.
- Overview videos.
- Algorithmic, auto-grading online homework for each section of the text. Most problems have video help tied to the question.
- A collection of printable resources created by Shana Calaway for the Open Course Library project.

How is Applied Calculus Different?

Students who plan to go into science, engineering, or mathematics take a year-long sequence of classes that cover many of the same topics as we do in our one-quarter or one-semester course. Here are some of the differences:

No trigonometry

We will not be using trigonometry at all in this course. The scientists and engineers need trigonometry frequently, and so a great deal of the engineering calculus course is devoted to trigonometric functions and the situations they can model.

The applications are different

The scientists and engineers learn how to apply calculus to physics problems, such as work. They do a lot of geometric applications, like finding minimum distances, volumes of revolution, or arclengths. In this class, we will do only a few of these (distance/velocity problems, areas between curves). On the other hand, we will learn to apply calculus in some economic and business settings, like maximizing profit or minimizing average cost, finding elasticity of demand, or finding the present value of a continuous income stream. Additionally we will apply calculus in life and social science settings, like determining the rate at which drug concentration in the body is changing, or exploring the rate at which a subject learns. These are applications that are seldom seen in a course for engineers.

Fewer theorems, no proofs

The focus of this course is applications rather than theory. In this course, we will use the results of some theorems, but we won't prove any of them. When you finish this course, you should be able to solve many kinds of problems using calculus, but you won't be prepared to go on to higher mathematics.

Less algebra

In this class, you will not need clever algebra. If you need to solve an equation, it will either be relatively simple, or you can use technology to solve it. In most cases, you won't need "exact answers;" calculator numbers will be good enough.

Simplification and Calculator Numbers

When you were in tenth grade, your math teacher may have impressed you with the need to simplify your answers. I'm here to tell you – she was wrong. The form your answer should be in depends entirely on what you will do with it next. In addition, the process of "simplifying," often messy algebra can ruin perfectly correct answers. From the teacher's point of view, "simplifying" obscures how a student arrived at his answer, and makes problems harder to grade. Moral: don't spend a lot of extra time simplifying your answer. Leave it as close to how you arrived at it as possible.

When should you simplify?

1. Simplify when it actually makes your life easier. For example, in Chapter 2 it's easier to find a second derivative if you simplify the first derivative.

2. Simplify your answer when you need to match it to an answer in the book. You may need to do some algebra to be sure your answer and the book answer are the same.

When you use your calculator

A calculator is required for this course, and it can be a wonderful tool. However, you should be careful not to rely too strongly on your calculator. Follow these rules of thumb:

1. Estimate your answers. If you expect an answer of about 4, and your calculator says 2500, you've made an error somewhere.
2. Don't round until the very end. Every time you make a calculation with a rounded number, your answer gets a little bit worse.
3. When you answer an applied problem, find a calculator number. It doesn't mean much to suggest that the company should produce $\frac{\sqrt{12100(2.4)}}{2.5}$ items; it's much more meaningful to report that they should produce about 106 items.
4. When you present your final answer, round it to something that makes sense. If you've found an amount of US money, round it to the nearest cent. If you've computed the number of people, round to the nearest person. If there's no obvious context, show your teacher at least two digits after the decimal place.
5. Occasionally in this course, you will need to find the "exact answer." That means – not a calculator approximation. (You can still use your calculator to check your answer.)

Licensing

A detailed breakdown of this resource's licensing can be found in ***Back Matter/Detailed Licensing***.

Note on Butler CC Edition

Note on Butler CC Remix

This content is a slight remix of the book *Applied Calculus*, by Calaway, Hoffman and Lippman. In the standard LibreText edition, the exercise sets for all sections of a given chapter appeared in a single, combined page at the end of the chapter. This remix moves the exercise sets for each individual section to a separate page, which appears at the end of the relevant section. Additionally, some small corrections/clarifications have been added to some exercises.

CHAPTER OVERVIEW

1: Review

1.1: Functions

What is a Function?

The natural world is full of relationships between quantities that change. When we see these relationships, it is natural for us to ask “If I know one quantity, can I then determine the other?” This establishes the idea of an input quantity, or independent variable, and a corresponding output quantity, or dependent variable. From this we get the notion of a functional relationship in which the output can be determined from the input.

For some quantities, like height and age, there are certainly relationships between these quantities. Given a specific person and any age, it is easy enough to determine their height, but if we tried to reverse that relationship and determine height from a given age that would be problematic, since most people maintain the same height for many years.

Function

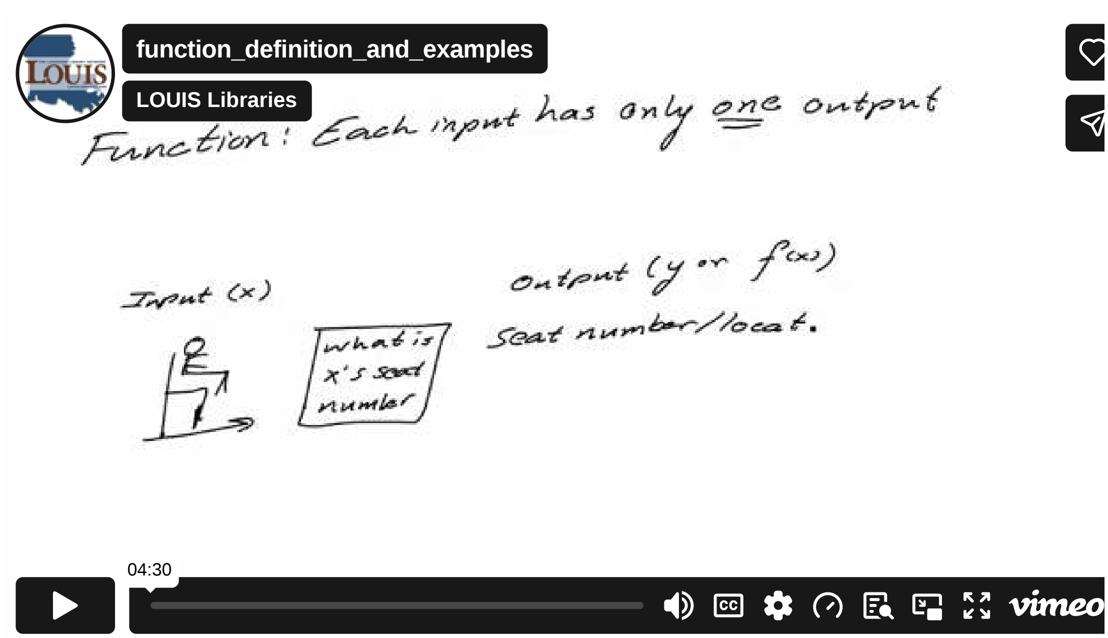
A **function** is rule for a relationship between an input, or independent, quantity and an output, or dependent, quantity in which each input value uniquely determines one output value. We say “the output is a function of the input.”

Example 1.1.1

In the height and age example above, is height a function of age? Is age a function of height?

In the height and age example above, it would be correct to say that height is a function of age, since each age uniquely determines a height. For example, on my 18th birthday, I had exactly one height of 69 inches.

However, age is not a function of height, since one height input might correspond with more than one output age. For example, for an input height of 70 inches, there is more than one output of age since I was 70 inches at the age of 20 and 21.



Function Notation

To simplify writing out expressions and equations involving functions, a simplified notation is often used. We also use descriptive variables to help us remember the meaning of the quantities in the problem.

Rather than write “height is a function of age,” we could use the descriptive variable h to represent height and we could use the descriptive variable a to represent age.

"height is a function of age"	if we name the function f we write
"h is f of a"	or more simply
$h = f(a)$	we could instead name the function h and write

"height is a function of age"	if we name the function f we write
$h(a)$	which is read "h of a"

Remember we can use any variable to name the function; the notation $h(a)$ shows us that h depends on a. The value "a" must be put into the function "h" to get a result. Be careful – the parentheses indicate that age is input into the function (Note: do not confuse these parentheses with multiplication!).

Function Notation

The notation output = f(input) defines a function named f. This would be read "output is f of input."

Example 1.1.2

A function $N = f(y)$ gives the number of police officers, N, in a town in year y. What does $f(2005) = 300$ tell us?

Solution

When we read $f(2005) = 300$, we see the input quantity is 2005, which is a value for the input quantity of the function, the year (y). The output value is 300, the number of police officers (N), a value for the output quantity. Remember $N = f(y)$. So this tells us that in the year 2005 there were 300 police officers in the town.

Tables as Functions

Functions can be represented in many ways: Words (as we did in the last few examples), tables of values, graphs, or formulas. Represented as a table, we are presented with a list of input and output values.

This table represents the age of children in years and their corresponding heights. While some tables show all the information we know about a function, this particular table represents just some of the data available for height and ages of children.

(input) a, age in years	4	5	6	7	8	9	10
(output) h, height in inches	40	42	44	47	50	52	54

Example 1.1.3

Which of these tables define a function (if any)?

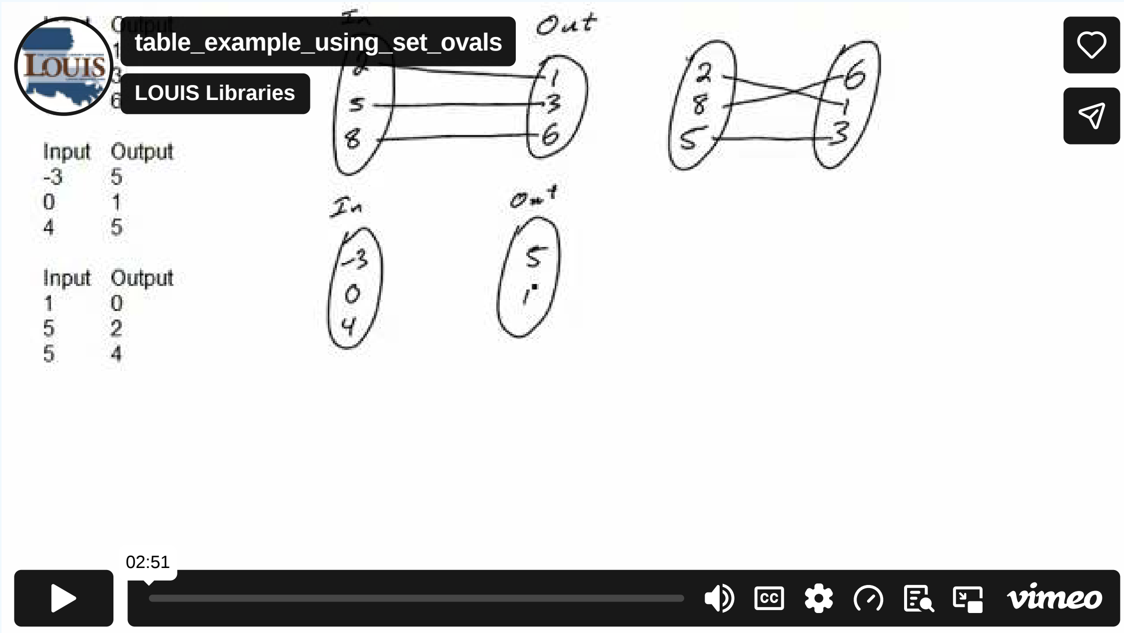
Input	Output
2	1
5	3
8	6

Input	Output
-3	5
0	1
4	5

Input	Output
1	0
5	2
5	

Solution

The first and second tables define functions. In both, each input corresponds to exactly one output. The third table does not define a function since the input value of 5 corresponds with two different output values.



Solving and Evaluating Functions

When we work with functions, there are two typical things we do: evaluate and solve. Evaluating a function is what we do when we know an input, and use the function to determine the corresponding output. Evaluating will always produce one result, since each input of a function corresponds to exactly one output.

Solving equations involving a function is what we do when we know an output, and use the function to determine the inputs that would produce that output. Solving a function could produce more than one solution, since different inputs can produce the same output.

Example 1.1.4

Using the table shown, where $Q = g(n)$

n	1	2	3	4	5
Q	8	6	7	6	8

a. Evaluate $g(3)$
b. Solve $g(n) = 6$

Solution

Evaluating g(3) (read: “g of 3”) means that we need to determine the output value, Q, of the function g given the input value of n=3. Looking at the table, we see the output corresponding to n=3 is Q=7, allowing us to conclude g(3) = 7.

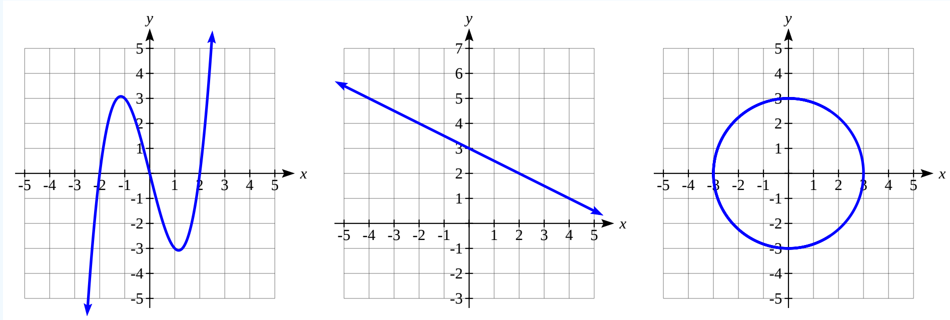
Solving g(n) = 6 means we need to determine what input values, n, produce an output value of 6. Looking at the table we see there are two solutions: n = 2 and n = 4. When we input 2 into the function g, our output is Q = 6 When we input 4 into the function g, our output is also Q = 6

Graphs as Functions

Oftentimes a graph of a relationship can be used to define a function. By convention, graphs are typically created with the input quantity along the horizontal axis and the output quantity along the vertical.

Example 1.1.5

Which of these graphs defines a function $y = f(x)$?



Solution

Looking at the three graphs above, the first two define a function y=f(x), since for each input value along the horizontal axis there is exactly one output value corresponding, determined by the y-value of the graph. The 3rd graph does not define a function y=f(x) since some input values, such as x=2, correspond with more than one output value.

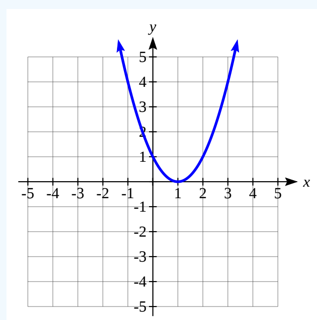
Vertical Line Test

The **vertical line test** is a handy way to think about whether a graph defines the vertical output as a function of the horizontal input. Imagine drawing vertical lines through the graph. If any vertical line would cross the graph more than once, then the graph does not define only one vertical output for each horizontal input.

Evaluating a function using a graph requires taking the given input and using the graph to look up the corresponding output. Solving a function equation using a graph requires taking the given output and looking on the graph to determine the corresponding input.

Example 1.1.6

Given the graph below,



a. Evaluate $f(2)$.
b. Solve $f(x) = 4$.

Solution

a) To evaluate f(2), we find the input of x=2 on the horizontal axis. Moving up to the graph gives the point (2, 1), giving an output of y=1. So f(2) = 1

b) To solve f(x) = 4, we find the value 4 on the vertical axis because if f(x) = 4 then 4 is the output. Moving horizontally across the graph gives two points with the output of 4: (-1,4) and (3,4). These give the two solutions to f(x) = 4: x = -1 or x = 3 This means f(-1)=4 and f(3)=4, or when the input is -1 or 3, the output is 4.

Notice that while the graph in the previous example is a function, getting two input values for the output value of 4 shows us that this function is not one-to-one.

Formulas as Functions

When possible, it is very convenient to define relationships using formulas. If it is possible to express the output as a formula involving the input quantity, then we can define a function.

Example 1.1.7

Express the relationship $2n+6p=12$ as a function $p=f(n)$ if possible.

Solution

To express the relationship in this form, we need to be able to write the relationship where p is a function of n, which means writing it as $p=$ [something involving n].

$$\begin{aligned} &2n+6p=12 && \text{subtract } 2n \text{ from both sides} \\ &6p=12-2n && \text{divide both sides by 6 and simplify} \\ &p=\tfrac{12-2n}{6}=\tfrac{12}{6}-\tfrac{2n}{6}=2-\tfrac{1}{3}n \end{aligned}$$

Having rewritten the formula as $p=$, we can now express p as a function:

$$p=f(n)=2-\frac{1}{3}n$$

Not every relationship can be expressed as a function with a formula. As with tables and graphs, it is common to evaluate and solve functions involving formulas. Evaluating will require replacing the input variable in the formula with the value provided and calculating. Solving will require replacing the output variable in the formula with the value provided, and solving for the input(s) that would produce that output.

Example 1.1.8

Given the function $k(t)=t^3+2$:

a. Evaluate $k(2)$.
b. Solve $k(t)=1$.

Solution

a) To evaluate $k(2)$, we plug in the input value 2 into the formula wherever we see the input variable t, then simplify

$$\begin{aligned} k(2)&=2^3+2 \\ k(2)&=8+2 \end{aligned}$$

So $k(2)=10$

b) To solve $k(t)=1$, we set the formula for $k(t)$ equal to 1 , and solve for the input value that will produce that output

$$k(t)=1 \quad \text{substitute the original formula } k(t)=t^3+2$$

$t^3+2=1$ subtract 2 from each side
$t^3=-1$ take the cube root of each side
$t=-1$

When solving an equation using formulas, you can check your answer by using your solution in the original equation to see if your calculated answer is correct.

We want to know if $k(t)=1$ is true when $t=-1$.

$$\begin{aligned} k(-1)&=(-1)^3+2 \\ &=-1+2 \\ &=1 \text{ which was the desired result.} \end{aligned} \tag{1.1.1}$$

Basic Toolkit Functions

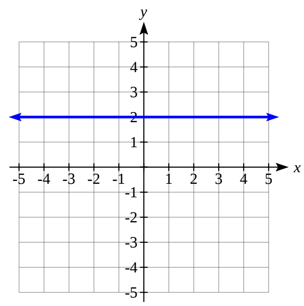
There are some basic functions that it is helpful to know the name and shape of. We call these the basic "toolkit of functions." For these definitions we will use x as the input variable and $f(x)$ as the output variable.

Toolkit Functions

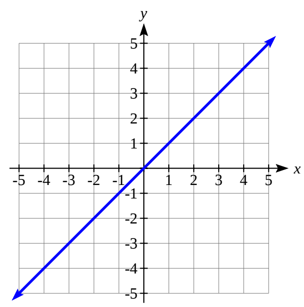
Linear	
Constant:	$f(x)=c$, where c is a constant (number)
Identity:	$f(x)=x$
Absolute Value:	$f(x)=\lvert x\rvert$
Power	

Quadratic:	$f(x) = x^2$
Cubic:	$f(x) = x^3$
Reciprocal:	$f(x) = \frac{1}{x}$
Reciprocal squared:	$f(x) = \frac{1}{x^2}$
Square root:	$f(x) = \sqrt[2]{x} = \sqrt{x}$
Cube root:	$f(x) = \sqrt[3]{x}$

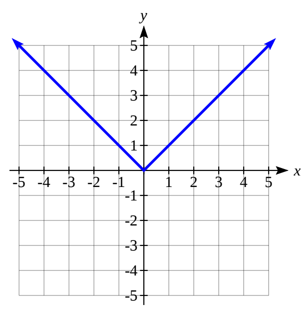
Graphs of the Toolkit Functions



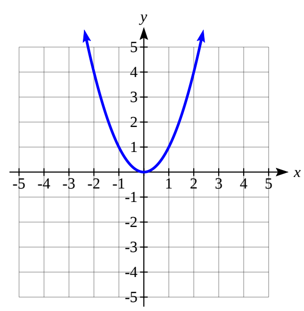
Constant: $f(x) = c$ (c is 2 in this picture).



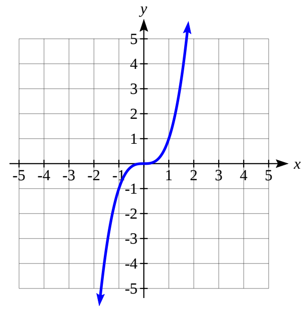
Identity: $f(x) = x$



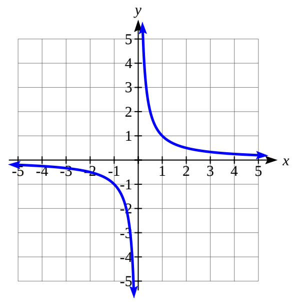
Absolute Value: $f(x) = |x|$



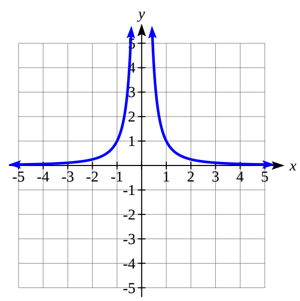
Quadratic: $f(x) = x^2$



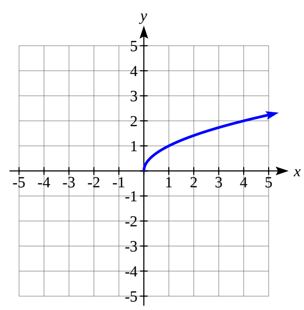
Cubic: $f(x)=x^3$



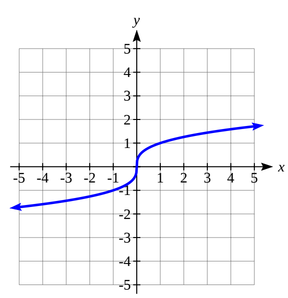
Reciprocal: $f(x)=\frac{1}{x}$



Reciprocal squared: $f(x)=\frac{1}{x^2}$



Square root: $f(x)=\sqrt[2]{x}=\sqrt{x}$



Cube root: $f(x)=\sqrt[3]{x}$

One of our main goals in mathematics is to model the real world with mathematical functions. In doing so, it is important to keep in mind the limitations of those models we create.

This table shows a relationship between circumference and height of a tree as it grows.

Circumference, c	1.7	2.5	5.5	8.2	13.7
Height, h	24.5	31	45.2	54.6	92.1

While there is a strong relationship between the two, it would certainly be ridiculous to talk about a tree with a circumference of -3 feet, or a height of 3000 feet. When we identify limitations on the inputs and outputs of a function, we are determining the domain and range of the function.

Domain and Range

- **Domain:** The set of possible input values to a function.
- **Range:** The set of possible output values of a function

Example 1.1.9

Using the tree table above, determine a reasonable domain and range.

Solution

We could combine the data provided with our own experiences and reason to approximate the domain and range of the function $h = f(c)$. For the domain, possible values for the input circumference c, it doesn't make sense to have negative values, so $c > 0$. We could make an educated guess at a maximum reasonable value, or look up that the maximum circumference measured is about 119 feet. With this information we would say a reasonable domain is feet.

Similarly for the range, it doesn't make sense to have negative heights, and the maximum height of a tree could be looked up to be 379 feet, so a reasonable range is feet.

A more compact alternative to inequality notation is **interval notation**, in which intervals of values are referred to by the starting and ending values. Curved parentheses are used for "strictly less than," and square brackets are used for "less than or equal to." Since infinity is not a number, we can't include it in the interval, so we always use curved parentheses with ∞ and $-\infty$. The table below will help you see how inequalities correspond to interval notation:

Inequality	Interval notation
$5 < h \leq 10$	(5, 10]
$5 \leq h < 10$	[5, 10)
$5 < h < 10$	(5, 10)
$h < 10$	$(-\infty, 10)$
$h \geq 10$	$[10, \infty)$
All real numbers ($\mathbb{R}$)	$(-\infty, \infty)$

Example 1.1.10

Describe the intervals of values shown on the line graph below using set builder and interval notations:

Solution

To describe the values, x, that lie in the intervals shown above we would say, "x is a real number greater than or equal to 1 and less than or equal to 3, or a real number greater than 5." As an inequality it is "$1 \leq x \leq 3$ or $x > 5$". In interval notation it is "$[1, 3] \cup (5, \infty)$".

Example 1.1.11

Find the domain of each function:

a. $f(x) = 2\sqrt{x+4}$
b. $g(x) = \dfrac{3}{6-3x}$

Solution

a. Since we cannot take the square root of a negative number, we need the inside of the square root to be non-negative. $x+4 \geq 0$ when $x \geq -4$, so the domain of $f(x)$ is $[-4, \infty)$.
b. We cannot divide by zero, so we need the denominator to be non-zero. $6-3x=0$ when $x=2$, so we must exclude 2 from the domain. The domain of $g(x)$ is $(-\infty, 2) \cup (2, \infty)$.

1.1.E: Exercises

1.1 Exercises

Exercise 1.1.E.1

The amount of garbage, G, produced by a city with population p is given by $G = f(p)$. G is measured in tons per week, and p is measured in thousands of people.

1. The town of Tola has a population of 40,000 and produces 13 tons of garbage each week. Express this information in terms of the function f.
2. Explain the meaning of the statement $f(5) = 2$.

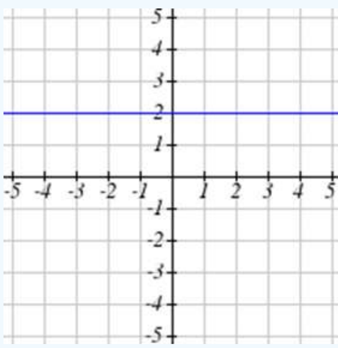
Exercise 1.1.E.2

The number of cubic yards of dirt, D, needed to cover a garden with area a square feet is given by $D = g(a)$.

1. A garden with area 5000 ft^2 requires 50 cubic yards of dirt. Express this information in terms of the function g.
2. Explain the meaning of the statement $g(100) = 1$.

Exercise 1.1.E.3

Select all of the following graphs which represent y as a function of x.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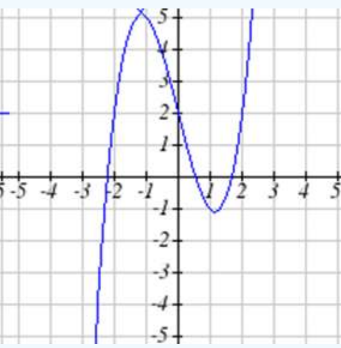
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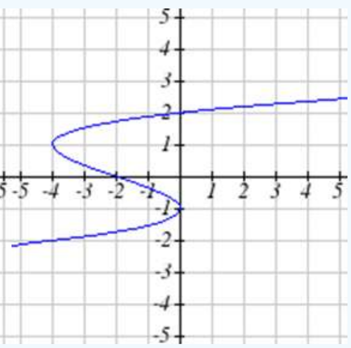
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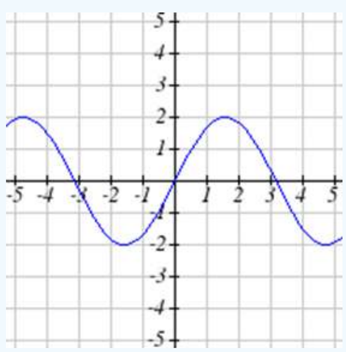
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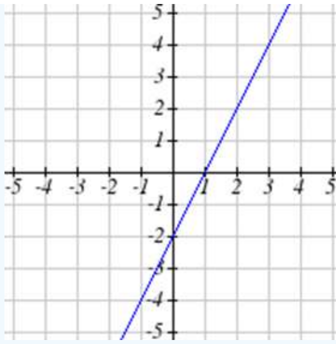

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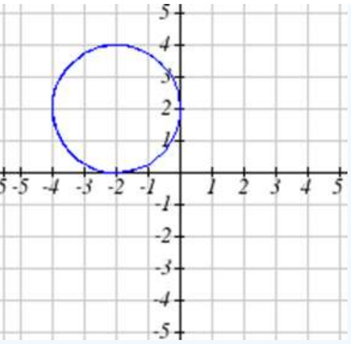

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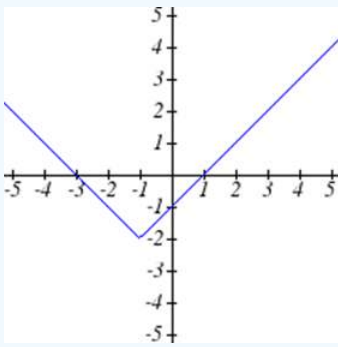
❓ Exercise $1.1.E.4$

Select all of the following graphs which represent y as a function of x.



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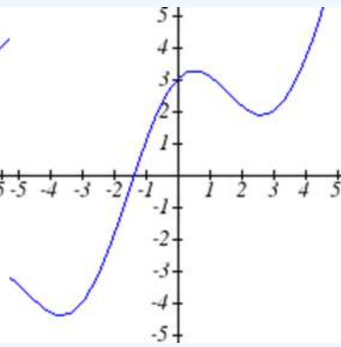

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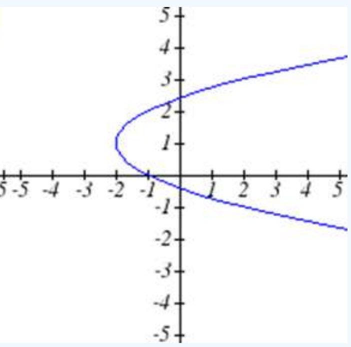

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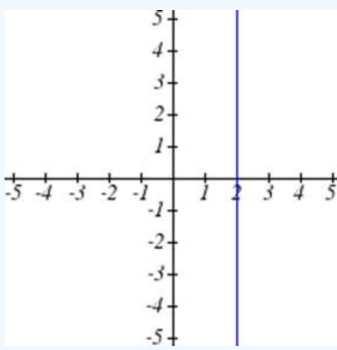

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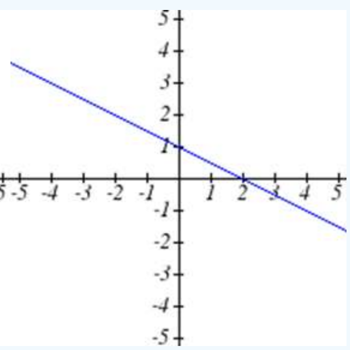

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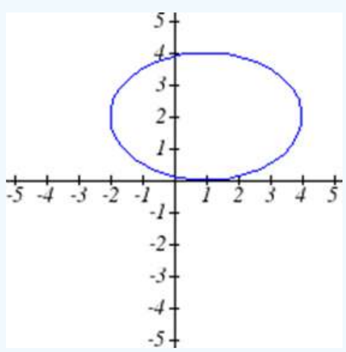


f.

? Exercise $1.1.E.5$

Select all of the following tables which represent y as a function of x.

a.

x	5	10	15
y	3	8	14

b.

x	5	10	15
y	3	8	8

c.

x	5	10	10
y	3	8	14

? Exercise $1.1.E.6$

Select all of the following tables which represent y as a function of x.

a.

x	2	6	13
y	3	10	10

b.

x	2	6	6
y	3	10	14

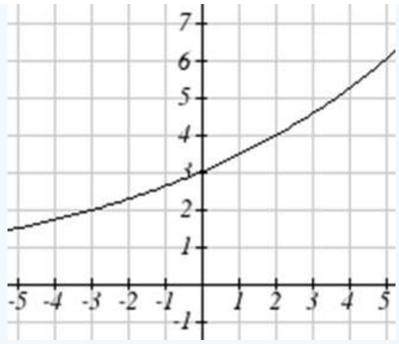
c.

x	2	6	13
y	3	10	14

? Exercise $1.1.E.7$

Given the function $g(x)$ graphed here,

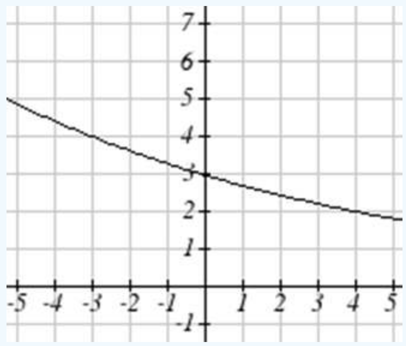
1. Evaluate $g(2)$
2. Solve $g(x) = 2$



? Exercise 1.1.E.8

Given the function $f(x)$ graphed here.

1. Evaluate $f(4)$
2. Solve $f(x) = 4$



? Exercise 1.1.E.9

Based on the table below,

1. Evaluate $f(3)$
2. Solve $f(x) = 1$

x	0	1	2	3	4	5	6	7	8	9
$f(x)$	74	28	1	53	56	3	36	45	14	47

? Exercise 1.1.E.10

Based on the table below,

1. Evaluate $f(8)$
2. Solve $f(x) = 7$

x	0	1	2	3	4	5	6	7	8	9
$f(x)$	62	8	7	38	86	73	70	39	75	34

? Exercises 1.1.E.11 – 1.1.E.18

For each of the following functions, evaluate: $f(-2)$, $f(-1)$, $f(0)$, $f(1)$, and $f(2)$

11. $f(x) = 4 - 2x$	12. $f(x) = 8 - 3x$

13. $f(x) = 8x^2 - 7x + 3$	14. $f(x) = 6x^2 - 7x + 4$
15. $f(x) = 3 + \sqrt{x+3}$	16. $f(x) = 4 - \sqrt[3]{x-2}$
17. $f(x) = \frac{x-3}{x+1}$	18. $f(x) = \frac{x-2}{x+2}$

Exercise 1.1.E.19

Let $f(t) = 3t + 5$

1. Evaluate $f(0)$
2. Solve $f(t) = 0$

Exercise 1.1.E.20

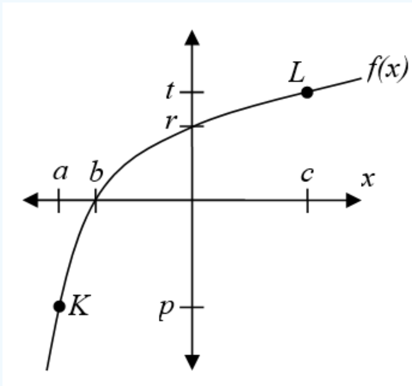
Let $g(p) = 6 - 2p$

1. Evaluate $g(0)$
2. Solve $g(p) = 0$

Exercise 1.1.E.21

Using the graph shown,

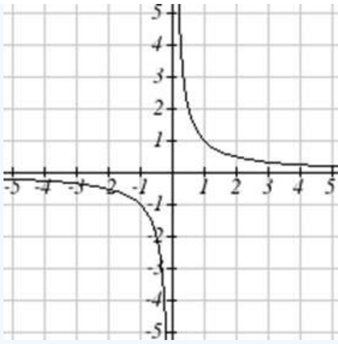
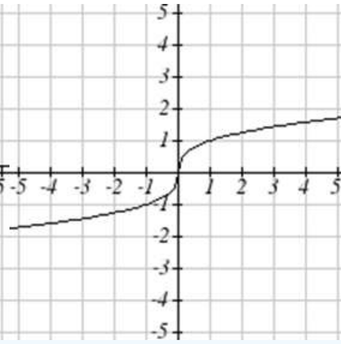
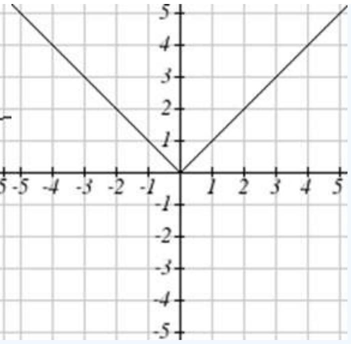
1. Evaluate $f(c)$
2. Solve $f(x) = p$
3. What are the coordinates of points L and K?



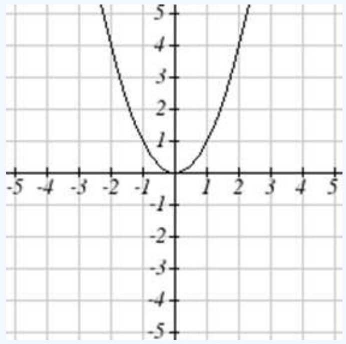
Exercise 1.1.E.22

Match each graph with its equation.

a. $y = x$	b. $y = x^3$	$y = \sqrt[3]{x}$	d. $y = \frac{1}{x}$
e. $y = x^2$	f. $y = \sqrt{x}$	g. $y = \|x\|$	h. $y = \frac{1}{x^2}$

i. ii. iii.



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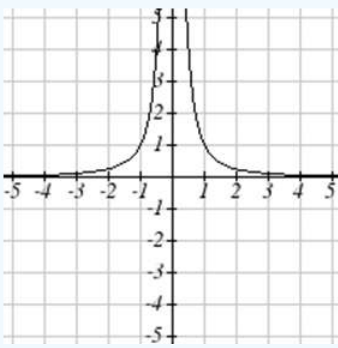
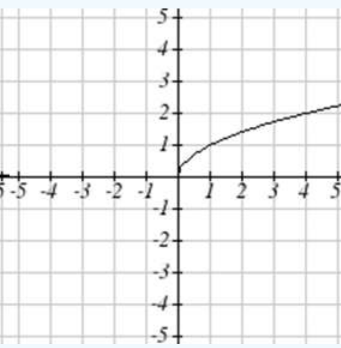
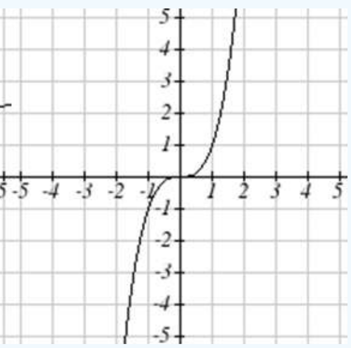
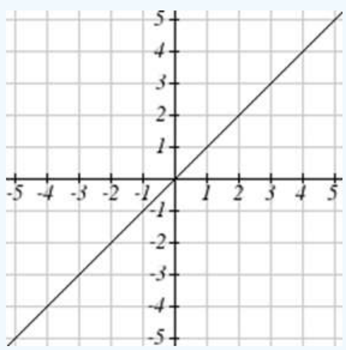




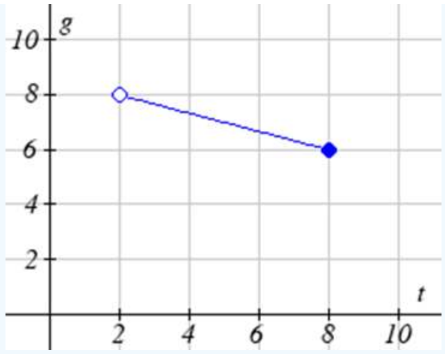
v. vi. vii.



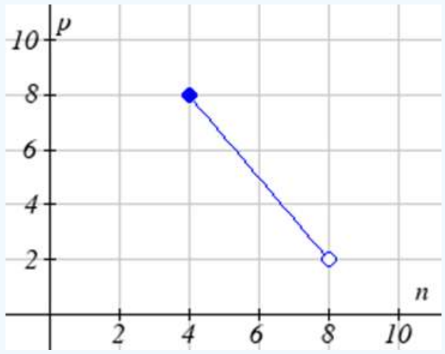
viii.

For Exercises $1.1.E.23 - 1.1.E.24$, write the domain and range of each graph as an inequality.

? Exercise $1.1.E.23$



Exercise 1.1.E.24



Exercise 1.1.E.25 – 1.1.E.30

Find the domain of each function.

25. $f(x) = 3\sqrt{x-2}$	26. $f(x) = 5\sqrt{x+3}$
27. $f(x) = \frac{9}{x-6}$	28. $f(x) = \frac{6}{x-8}$
29. $f(x) = \frac{3x+1}{4x+2}$	30. $f(x) = \frac{5x+3}{4x-1}$

1.2: Operations on Functions

Composition of Functions

Suppose we wanted to calculate how much it costs to heat a house on a particular day of the year. The cost to heat a house will depend on the average daily temperature, and the average daily temperature depends on the particular day of the year. Notice how we have just defined two relationships: The temperature depends on the day, and the cost depends on the temperature. Using descriptive variables, we can notate these two functions.

The first function, $C(T)$, gives the cost C of heating a house when the average daily temperature is T degrees Celsius, and the second, $T(d)$, gives the average daily temperature of a particular city on day d of the year. If we wanted to determine the cost of heating the house on the fifth day of the year, we could do this by linking our two functions together, an idea called composition of functions. Using the function $T(d)$, we could evaluate $T(5)$ to determine the average daily temperature on the fifth day of the year. We could then use that temperature as the input to the $C(T)$ function to find the cost to heat the house on the fifth day of the year: $C(T(5))$.

Composition of Functions

When the output of one function is used as the input of another, we call the entire operation a **composition of functions**. We write $f(g(x))$, and read this as "f of g of x" or "f composed with g at x".

An alternate notation for composition uses the composition operator: $\circ$. $(f \circ g)(x)$ is read "f of g of x" or "f composed with g at x", just like $f(g(x))$.

Example 1.2.1

Suppose $c(s)$ gives the number of calories burned doing s sit-ups, and $s(t)$ gives the number of sit-ups a person can do in t minutes. Interpret $c(s(3))$.

Solution

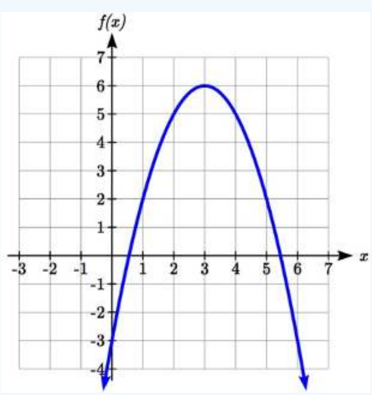
When we are asked to interpret, we are being asked to explain the meaning of the expression in words. The inside expression in the composition is $s(3)$. Since the input to the s function is time, the 3 is representing 3 minutes, and $s(3)$ is the number of sit-ups that can be done in 3 minutes. Taking this output and using it as the input to the $c(s)$ function will give us the calories that can be burned by the number of sit-ups that can be done in 3 minutes.

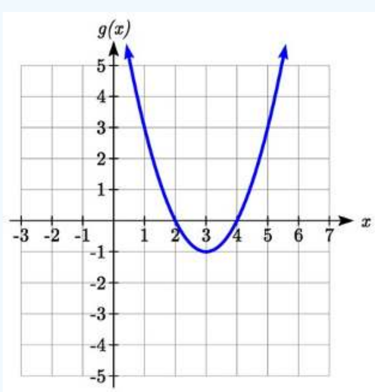
Composition of Functions using Tables and Graphs

When working with functions given as tables and graphs, we can look up values for the functions using a provided table or graph. We start evaluation from the provided input, and first evaluate the inside function. We can then use the output of the inside function as the input to the outside function. To remember this, always work from the inside out.

Example 1.2.2

Using the graphs below, evaluate $f(g(1))$.





Solution

To evaluate $f(g(1))$, we again start with the inside evaluation. We evaluate $g(1)$ using the graph of the $g(x)$ function, finding the input of 1 on the horizontal axis and finding the output value of the graph at that input. Here, $g(1) = 3$. Using this value as the input to the f function, $f(g(1)) = f(3)$. We can then evaluate this by looking to the graph of the $f(x)$ function, finding the input of 3 on the horizontal axis, and reading the output value of the graph at this input. Here, $f(3) = 6$, so $f(g(1)) = 6$.

Compositions using Formulas

When evaluating a composition of functions where we have either created or been given formulas, the concept of working from the inside out remains the same. First we evaluate the inside function using the input value provided, then use the resulting output as the input to the outside function.

✓ Example 1.2.3

Given $f(t) = t^2 - t$ and $h(x) = 3x + 2$, evaluate $f(h(1))$.

Solution

Since the inside evaluation is $h(1)$ we start by evaluating the $h(x)$ function at 1:

$$h(1) = 3(1) + 2 = 5$$

Then $f(h(1)) = f(5)$, so we evaluate the $f(t)$ function at an input of 5:

$$f(h(1)) = f(5) = 5^2 - 5 = 20$$

We are not limited, however, to using a numerical value as the input to the function. We can put anything into the function: a value, a different variable, or even an algebraic expression, provided we use the input expression everywhere we see the input variable.

✓ Example 1.2.4

Let $f(x) = x^2$ and $g(x) = \dfrac{1}{x} - 2x$. Find $f(g(x))$ and $g(f(x))$.

Solution

To find $f(g(x))$, we start by evaluating the inside, writing out the formula for $g(x)$:

$$g(x) = \frac{1}{x} - 2x$$

We then use the expression $\left(\dfrac{1}{x} - 2x\right)$ as the input for the function f:

$$f(g(x)) = f\left(\frac{1}{x} - 2x\right)$$

We then evaluate the function $f(x)$ using the formula for $g(x)$ as the input. Since $f(x) = x^2$ then

$$f\left(\frac{1}{x} - 2x\right) = \left(\frac{1}{x} - 2x\right)^2$$

This gives us the formula for the composition:

$$f(g(x)) = \left(\frac{1}{x} - 2x\right)^2$$

Likewise, to find $g(f(x))$, we evaluate the inside, writing out the formula for $f(x)$: $g(f(x)) = g(x^2)$.

Now we evaluate the function $g(x)$ using x^2 as the input:

$$g(f(x)) = \frac{1}{x^2} - 2x^2$$

Example 1.2.5

A city manager determines that the tax revenue, R, in millions of dollars collected on a population of p thousand people is given by the formula $R(p) = 0.03p + \sqrt{p}$, and that the city's population, in thousands, is predicted to follow the formula $p(t) = 60 + 2t + 0.3t^2$, where t is measured in years after 2010. Find a formula for the tax revenue as a function of the year.

Solution

Since we want tax revenue as a function of the year, we want year to be our initial input, and revenue to be our final output. To find revenue, we will first have to predict the city population, and then use that result as the input to the tax function. So we need to find $R(p(t))$. Evaluating this,

$$\begin{aligned} R(p(t)) &= R(60 + 2t + 0.3t^2) \\ &= 0.03(60 + 2t + 0.3t^2) + \sqrt{60 + 2t + 0.3t^2} \end{aligned}$$

This composition gives us a single formula which can be used to predict the tax revenue during a given year, without needing to find the intermediary population value.

For example, to predict the tax revenue in 2017, when $t = 7$ (because t is measured in years after 2010),

$$\begin{aligned} R(p(7)) &= 0.03\left(60 + 2(7) + 0.3\left(7^2\right)\right) + \sqrt{60 + 2(7) + 0.3\left(7^2\right)} \\ &\approx 12.079 \text{ million dollars} \end{aligned}$$

Later in this course, it will be desirable to decompose a function – to write it as a composition of two simpler functions.

Example 1.2.6

Write $f(x) = 3 + \sqrt{5 - x^2}$ as the composition of two functions.

Solution

We are looking for two functions, g and h, so $f(x) = g(h(x))$. To do this, we look for a function inside a function in the formula for $f(x)$. As one possibility, we might notice that $5 - x^2$ is the inside of the square root. We could then decompose the function as:

$$h(x) = 5 - x^2, \quad g(x) = 3 + \sqrt{x}$$

We can check our answer by recomposing the functions:

$$g(h(x)) = g(5 - x^2) = 3 + \sqrt{5 - x^2}$$

Note that this is not the only solution to the problem. Another non-trivial decomposition would be

$$h(x) = x^2, \quad g(x) = 3 + \sqrt{5 - x}.$$

Transformations of Functions

Transformations allow us to construct new equations from our basic toolkit functions. The most basic transformations are shifting the graph vertically or horizontally.

Vertical Shift

Given a function $f(x)$, if we define a new function $g(x)$ as $g(x) = f(x) + k$, where k is a constant, then $g(x)$ is a **vertical shift** of the function $f(x)$, where all the output values have been increased by k.

If k is positive, then the graph will shift up. If k is negative, then the graph will shift down.

Horizontal Shift

Given a function $f(x)$, if we define a new function $g(x)$ as $g(x) = f(x+k)$, where k is a constant, then $g(x)$ is a **horizontal shift** of the function $f(x)$, where all the output values have been increased by k.

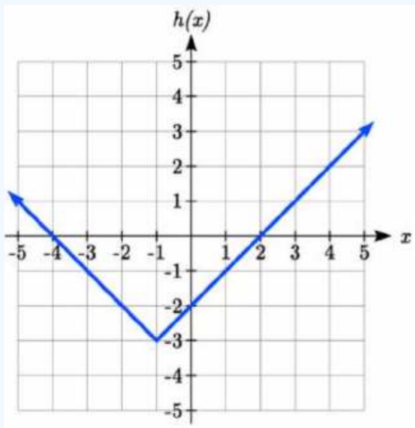
If k is positive, then the graph will shift left. If k is negative, then the graph will shift right.

Example 1.2.7

Given $f(x) = |x|$, sketch a graph of $h(x) = f(x+1) - 3 = |x+1| - 3$.

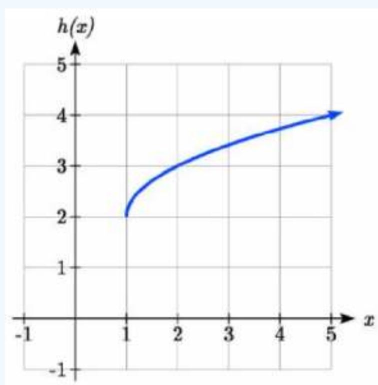
Solution

The function f is our toolkit absolute value function. We know that this graph has a V shape, with the point at the origin. The graph of h has transformed f in two ways: $f(x+1)$ is a change on the inside of the function, giving a horizontal shift left by 1, then the subtraction by 3 in $f(x+1) - 3$ is a change to the outside of the function, giving a vertical shift down by 3. Transforming the graph gives:



Example 1.2.8

Write a formula for the graph shown, a transformation of the toolkit square root function.



Solution

The graph of the toolkit function starts at the origin, so this graph has been shifted 1 to the right, and up 2. In function notation, we could write that as $h(x) = f(x-1) + 2$. Using the formula for the square root function we can write $h(x) = \sqrt{x-1} + 2$.

Note that this transformation has changed the domain and range of the function. This new graph has domain $[1, \infty)$ and range $[2, \infty)$.

Another transformation that can be applied to a function is a reflection over the horizontal or vertical axis.

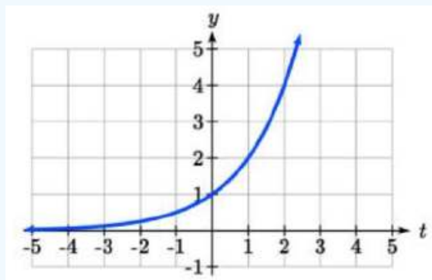
Reflections

Given a function $f(x)$, if we define a new function $g(x)$ as $-f(x)$, then $g(x)$ is a **vertical reflection** of the function $f(x)$, sometimes called a reflection about the x-axis

If we define a new function $g(x)$ as $f(-x)$, then $g(x)$ is a **horizontal reflection** of the function $f(x)$, sometimes called a reflection about the y-axis.

Example 1.2.9

A common model for learning has an equation similar to $k(t) = -2^t + 1$, where k is the percentage of mastery that can be achieved after t practice sessions. This is a transformation of the function $f(t) = 2^t$ shown here. Sketch a graph of $k(t)$.



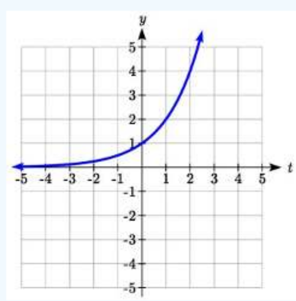
$f(t) = 2^t$

Solution

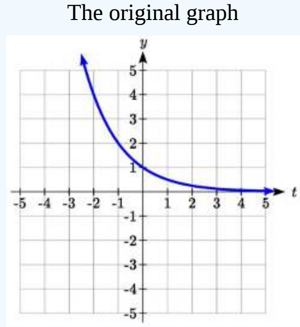
This equation combines three transformations into one equation.

A horizontal reflection:	$f(-t) = 2^{-t}$	combined with
a vertical reflection:	$-f(-t) = -2^{-t}$	combined with
a vertical shift up 1:	$-f(-t) + 1 = -2^{-t} + 1.$	

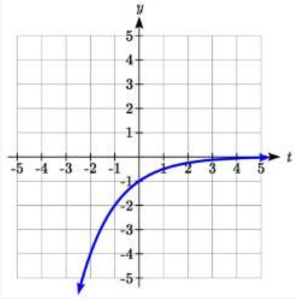
We can sketch a graph by applying these transformations one at a time to the original function:



The original graph



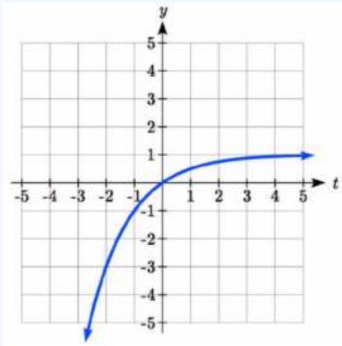
Horizontally reflected



Then vertically reflected

Then, after shifting up 1, we get the final graph:

$$k(t) = -f(-t) + 1 = -2^{-t} + 1$$



Final graph

Note: As a model for learning, this function would be limited to a domain of $t \geq 0$, with corresponding range $[0, 1)$.

With shifts, we saw the effect of adding or subtracting to the inputs or outputs of a function. We now explore the effects of multiplying the outputs.

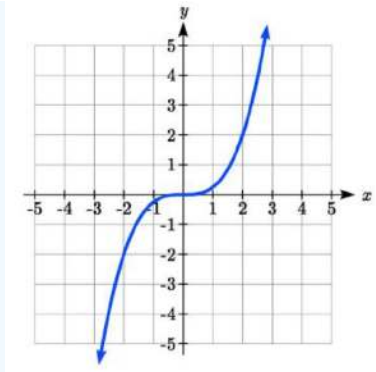
Vertical Stretch/Compression

Given a function $f(x)$, if we define a new function $g(x)$ as $g(x) = k \cdot f(x)$, where k is a constant, then $g(x)$ is a **vertical stretch or compression** of the function $f(x)$.

- If $k > 1$, then the graph will be stretched
- If $0 < k < 1$, then the graph will be compressed
- If $k < 0$, then there will be combination of a vertical stretch or compression with a vertical reflection

Example $1.2.10$

The graph below is a transformation of the toolkit function $f(x) = x^3$. Relate this new function $g(x)$ to $f(x)$, then find a formula for $g(x)$.



Solution

When trying to determine a vertical stretch or shift, it is helpful to look for a point on the graph that is relatively clear. In this graph, it appears that $g(2) = 2$. With the basic cubic function at the same input, $f(2) = 2^3 = 8$. Based on that, it appears that the outputs of g are $\frac{1}{4}$ the outputs of the function f, since $g(2) = \frac{1}{4} f(x)$. From this we can fairly safely conclude that

$$g(x) = \frac{1}{4} f(x).$$

We can write a formula for g by using the definition of the function f:

$$g(x) = \frac{1}{4} f(x) = \frac{1}{4} x^3.$$

Combining Transformations

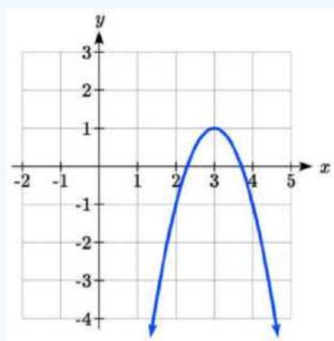
When combining vertical transformations, it is very important to consider the order of the transformations. For example, vertically shifting by 3 and then vertically stretching by 2 does not create the same graph as vertically stretching by 2 and then vertically shifting by 3. The order follows nicely from order of operations.

Combining Vertical Transformations

When combining vertical transformations written in the form $a \cdot f(x) + k$, first vertically stretch by a, then vertically shift by k.

Example 1.2.11

Write an equation for the transformed graph of the quadratic function shown.



Solution

Since this is a quadratic function, first consider what the basic quadratic tool kit function looks like and how this has changed. Observing the graph, we notice several transformations: The original tool kit function has been flipped over the x axis, some kind of stretch or compression has occurred, and we can see a shift to the right 3 units and a shift up 1 unit. In total there are four operations:

1. Vertical Reflection, requiring a negative sign outside the function.
2. Vertical Stretch

3. Horizontal Shift Right 3 units, which tells us to put x-3 on the inside of the function.
4. Vertical Shift up 1 unit, telling us to add 1 on the outside of the function.

By observation, the basic tool kit function has a vertex at (0, 0) and symmetrical points at (1, 1) and (-1, 1). These points are 1 unit up and 1 unit over from the vertex. The new points on the transformed graph are 1 unit away horizontally but 2 units away vertically. They have been stretched vertically by 2.

Not everyone can see this by simply looking at the graph. If you can, great, but if not we can solve for it. First, we will write the equation for this graph, with an unknown vertical stretch:

$f(x)=x^2$	The original function
$-f(x)=-x^2$	Vertically reflected
$-a\cdot f(x)=-ax^2$	Vertically stretched
$-a\cdot f(x-3)=-a(x-3)^2$	Shifted right 3
$-a\cdot f(x-3)+1=-a(x-3)^2+1$	Shifted up 1

We now know our graph is going to have an equation of the form $g(x)=-a(x-3)^2+1$. To find the vertical stretch, we can identify any point on the graph (other than the highest point), such as the point (2,-1), which tells us $g(2)=-1$. Using our general formula, and substituting 2 for x, and -1 for $g(x)$.

This tells us that to produce the graph we need a vertical stretch by two.

Thus the function that produces this graph is

$$g(x)=-2(x-3)^2+1.$$

Example 1.2.12

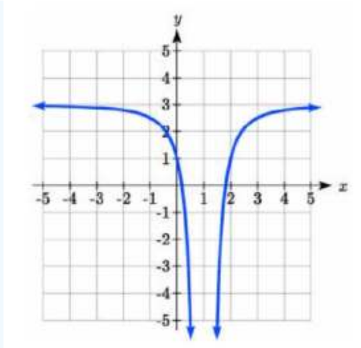
On what interval(s) is the function $g(x)=\dfrac{-2}{(x-1)^2}+3$ increasing and decreasing?

Solution

This is a transformation of the toolkit reciprocal squared function, $f(x)=\dfrac{1}{x^2}$

$-2f(x)=\dfrac{-2}{x^2}$	A vertical flip and vertical stretch by 2
$-2f(x-1)=\dfrac{-2}{(x-1)^2}$	A shift right by 1
$-2f(x-1)+3=\dfrac{-2}{(x-1)^2}+3$	A shift up by 3

The basic reciprocal squared function is increasing on $(-\infty,0)$ and decreasing on $(0,\infty)$. Because of the vertical flip, the $g(x)$ function will be decreasing on the left and increasing on the right. The horizontal shift right by 1 will also shift these intervals to the right one. From this, we can determine $g(x)$ will be increasing on $(1,\infty)$ and decreasing on $(-\infty,1)$. We also could graph the transformation to help us determine these intervals.



Graph of the transformed function.